

Project of MO434

A Comparative Study of Knowledge Distillation Schemes from Pretrained Image Classifiers into a Lightweight ConvNet

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Modern image classifiers — VGG [1], ResNet [2], ConvNeXt [3] — share a common macro-structure: an *encoder* that produces a spatial feature map, a *global average pooling* (GAP) that collapses spatial dimensions, and a linear *classifier*. Knowledge distillation Train the teacher’s classifier with its encoder frozen. Fine-tuning may be used, but it must preserve the encoder’s ImageNet knowledge. [4, 5] exploits this structure by training a smaller student to imitate intermediate representations of the pretrained teacher on ImageNet. The original distillation formulation of [4] matches the logits of the teacher and student through a softened cross-entropy. FitNets [5] extended this idea to intermediate feature maps using hint layers. Attention transfer [6] aligns spatial attention maps, while more recent methods, such as relational knowledge distillation [7], match pairwise relations among samples.

The setup of this project is closest to FitNets (feature-map variant) and to logit-free representation distillation (pooled-output variant), with an additional cross-entropy term on the student’s representation to keep the predictions aligned with the labels. The project involves three phases: (1) train the teacher’s classifier with its encoder frozen. Fine-tuning may be used, but it must preserve the encoder’s ImageNet knowledge; (2) train the student’s encoder+predictor using the same pretrained classifier of phase 1; and (3) test the student’s model with the same pretrained classifier of phase 1.

By forming groups of at most two people, the groups should address the following questions using multiple datasets (e.g., at least two) of their choice for image classification:

1. Which teacher transfers best?

Different backbones learn different features; the ease of imitation and the final classification accuracy may depend strongly on the teacher.

2. What should the student predict?

The pre-GAP feature maps ($7 \times 7 \times C$) or the post-GAP pooled vector (C -dimensional)? Note that the student’s encoder is followed by a predictor, whose architecture could use convolutional blocks to predict the teacher’s pre-GAP feature maps or dense blocks to predict the post-GAP pooled vector.

3. What are the best architectures for the student’s encoder + predictor?

The final model should considerably save GFLOPS and parameters as compared to the teacher.

4. What should we use as a loss function?

You may combine the MSE of the predictor and the cross-entropy of the classifier.

5. **What can you learn and use in this project from the literature of knowledge distillation?**

You may compare and/or adapt your models to the scheme proposed in [7].

The report must be written in Letter format, using 11pt font, and should include figures, tables, graphs, and references. It must follow the organization below:

- Cover page: include the course name, student's names in the group, academic identification number (RA) of each group member, and submission date, followed by a summary of the techniques implemented and the main results.
- Subsequent pages: present the literature studied for the project implementation, the difficulties encountered, the proposed solutions (architectures of the compared models), and their quantitative and qualitative results on each dataset, with illustrations and discussion.
- Final page: present a conclusion about what was accomplished in the project and provide suggestions for improvement.

Finally, provide the notebooks/Python scripts with the best student architecture for each teacher. Send code and report in the classroom platform.

References

- [1] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. In *ICLR*, 2015.
- [2] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016.
- [3] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. In *CVPR*, 2022.
- [4] G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- [5] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio. FitNets: Hints for thin deep nets. In *ICLR*, 2015.
- [6] S. Zagoruyko and N. Komodakis. Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer. In *ICLR*, 2017.
- [7] W. Park, D. Kim, Y. Lu, and M. Cho. Relational knowledge distillation. In *CVPR*, 2019.